Recipe 2

FASTQ quality control

What does the recipe do

This recipe performs data quality control steps on data obtained from SRA.

Input: sequencing data accession number: `SRR519926`

Output:

1. Original sequencing data for the accession number
2. FASTQC reports of the original sequencing data
3. Quality and adapter trimmed sequencing data
4. FASTQC reports of the trimmed data

Tools

- fastq-dump to download sequencing reads
- trimmomatic to perform quality trimming
- fastqc to visualize FASTQ quality

Related book chapters:

1. **Accessing the Short Read Archive (SRA)**
2. **Quality control of sequencing data**

Methodology

Methodology:

1. Download data with `fastq-dump`
2. Run `fastqc` on the "raw" sequencing reads
3. Create an adapter file that matches the sequence used library construction.
4. Run `trimmomatic` to trim reads by their quality and to remove adapter sequences
5. Run `fastqc` again this time on the trimmed data

Design choices

The recipe will distribute the data into separate directories `reads` and `reports`

For different library preparations you may need to change the adapter sequence. For example an Nextera type RNA-Seq may require:

```
echo ">nextera" > adapter.fa
echo "CTGTCTCTTATACACATCTCCGAGCCCACGAGAC" >> adapter.fa
```

Subtle effects

The order of quality control steps matters. Should you trim by quality or by adapter first?

```
ILLUMINACLIP:adapter.fa:2:30:5 SLIDINGWINDOW:4:20
```

will not produce the exact same output as

```
SLIDINGWINDOW:4:20 ILLUMINACLIP:adapter.fa:2:30:5
```

Think about the tradeoffs.

Where to go next

Make the script take parameters from command line: `SRR=$1` so that you can use it as:

```
bash recipe.sh SRR519926
```

Now factor out quality filtering parameters

```
  # ... SLIDINGWINDOW:4:$2 ...
```

so now you can use it as (are the results different?):

```
bash recipe.sh SRR519926 10
bash recipe.sh SRR519926 20
bash recipe.sh SRR519926 30
```

Troubleshooting

If you pass `fastq-dump` a non-existing name you will get a super confusing error message:

```
fastq-dump foo
```

will produce:

> *2018-10-01T15:14:21 fastq-dump.2.8.2 err: name not found while resolving query within virtual file system module - failed to resolve accession 'foo' - no data (404)*

gee thanks ... NCBI...

That's it!

Bon Appétit! 🍞